LINK LAYER DISCOVERY PROTOCOL (LLDP)

Link layer discovery Protocol (LLDP) is an open-standard layer 2 discovery protocol used by network devices to share basic information with their directly connected neighbors. Unlike Cisco ~~Device~~ Discovery Protocol (CDP), which works only on Cisco equipment, LLDP can operate across devices from different vendors. It allows switches, routers, phones, and other networked devices to advertise details such as their identity, port information and system capabilities.

LLDP works by/through periodic advertisements sent by each device on its active interfaces. These LLDP frames contain information such as the device's name, port ID, and system capabilities. Any directly connected neighbor ~~recieves~~ receives these frames and store the details in an LLDP neighbor table. Because LLDP is an open standard, this exchange work across device from different vendors, not just those from cisco.

When you run the commands like "show lldp neighbors", the device displays the information it has learned from these received advertisement. LLDP only shares data with directly connected devices, ensuring accurate physical-layer visibility. By continuously sending and listening for these small updates, LLDP helps map connection, verify cabling and understand how devices are linked in a mixed-vendor network.